

Total No. of Questions : 8]

SEAT No. :

PC1976

[Total No. of Pages : 2

[6353]-311

T.E. (Mechanical/Electrical)

HONORS IN ELECTRIC VEHICLES

E- Vehicle System Design

(2019 Pattern) (Semester - II) (302033MJ)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Four questions from the following.
- 2) Draw neat labeled diagrams wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of non-programmable electronic calculator is permitted.
- 5) Assume Suitable/Standard data jf necessary.

Q1) a) Explain with neat diagram topology of tyres design and its different component. [8]

b) Explain Topology design of wheels. [9]

OR

Q2) a) Explain elements of vehicle dynamics system. [8]

b) Explain vehicle and body centre of gravity for movement design of e- vehicles. [9]

Q3) a) Explain Powertrain in e-vehicle. [8]

b) Explain three wheel drive layout design. [9]

OR

Q4) a) Explain Transmission system in e- Vehicle. [8]

b) What is differential system and mention its types. [9]

P.T.O.

- Q5)** a) Explain heat shrink tubing used in battery assembly. [9]
b) Classify different geometry for assembling batteries in battery compartment.[9]

OR

- Q6)** a) Explain Battery compartment. [9]
b) Explain with neat diagram Liquid cooling used for battery compartment.[9]

- Q7)** a) Explain vehicle dynamics of EV. [9]
b) Explain crash analysis of EV. [9]

OR

- Q8)** a) Explain ergonomic design of roll cage frame. [9]
b) How optimization technique used in ergonomics design in EV. [9]

